



Prevalence and population attributable fractions of potentially modifiable risk factors for dementia in Canada: A cross-sectional analysis of the Canadian Longitudinal Study on Aging

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Abstract

Objectives We investigated the prevalence and population attributable fraction (PAF) of 12 potentially modifiable risk factors for dementia in middle-aged and older Canadians.

Methods We conducted a cross-sectional study of 30,097 adults aged 45 to 85 with baseline data from the Canadian Longitudinal Study on Aging (2011–2015). Risk factors and associated relative risks were taken from a highly cited systematic review. We calculated the prevalence of each risk factor using sampling weights. Individual PAFs were calculated both crudely and weighted for communality, and combined PAFs were calculated using both multiplicative and additive assumptions. Analyses were stratified by household income and repeated at CLSA's first follow-up (2015–2018).

Results The most prevalent risk factors were physical inactivity (63.8%; 95% CI, 62.8–64.9), hypertension (32.8%; 31.7–33.8), and obesity (30.8%; 29.7–31.8). The highest crude PAFs were physical inactivity (19.9%), traumatic brain injury (16.7%), and hypertension (16.6%). The highest weighted PAFs were physical inactivity (11.6%), depression (7.7%), and hypertension (6.0%). We estimated that the 12 risk factors combined accounted for 43.4% (37.3–49.0) of dementia cases assuming weighted multiplicative interactions and 60.9% (55.7–65.5) assuming additive interactions. There was a clear gradient of increasing prevalence and PAF with decreasing income for 9 of the 12 risk factors.

Conclusion The findings of this study can inform individual- and population-level dementia prevention strategies in Canada. Differences in the impact of individual risk factors between this study and other international and regional studies highlight the importance of tailoring national dementia strategies to the local distribution of risk factors.

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Résumé

Objectifs Nous avons étudié la prévalence et la fraction attribuable dans la population (FAP) de 12 facteurs de risque de démence potentiellement modifiables chez les Canadiens d'âge moyen et plus âgés.

Méthode Nous avons mené une étude transversale de 30 097 adultes de 45 à 85 ans à l'aide des données de référence de l'Étude longitudinale canadienne sur le vieillissement (ELCV) (2011–2015). Les facteurs de risque et les risques relatifs associés ont été extraits d'une revue systématique fréquemment citée. Nous avons calculé la prévalence de chaque facteur de risque à l'aide de poids d'échantillonnage. Les FAP individuelles ont été calculées à la fois sous forme brute et pondérées selon leurs points communs; les FAP combinées ont été calculées à l'aide d'hypothèses multiplicatives et additives. Les analyses ont été stratifiées selon le revenu du ménage et répétées au premier suivi de l'ELCV (2015–2018).

Résultats Les facteurs de risque les plus prévalents étaient la sédentarité (63,8 %; IC de 95%, 62,8–64,9), l'hypertension artérielle (32,8 %; 31,7–33,8) et l'obésité (30,8 %; 29,7–31,8). Les FAP brutes les plus élevées étaient la sédentarité (19,9 %), les traumatismes cranio-cérébraux (16,7 %) et l'hypertension artérielle (16,6 %). Les FAP pondérées les plus élevées étaient la sédentarité (11,6 %), la dépression (7,7 %) et l'hypertension artérielle (6,0 %). Selon nos estimations, les 12 facteurs de risque combinés représentaient 43,4 % (37,3–49,0) des cas de démence en supposant des interactions multiplicatives pondérées et 60,9 % (55,7–65,5) en supposant des interactions additives. Il y avait clairement un gradient d'accroissement de la prévalence et de la FAP avec la diminution du revenu pour 9 des 12 facteurs de risque.

Conclusion Les constats de l'étude peuvent éclairer les stratégies individuelles et populationnelles de prévention de la démence au Canada. Les différences d'impact des facteurs de risque individuels entre cette étude et d'autres études internationales et régionales montrent l'importance d'adapter les stratégies nationales de prévention de la démence à la répartition locale des facteurs de risque.

Keywords Dementia · Risk factors · Population health · Social determinants of health

Mots-clés Démence · facteurs de risque · santé de la population · déterminants sociaux de la santé

Introduction

Dementia is a clinical syndrome marked by typically progressive cognitive decline, affecting memory, perceptual-motor skills, language, mood, behaviour, and the ability to perform activities of daily living (American Psychiatric Association, 2022; WHO, 2023). In Canada, the number of people living with dementia was estimated to be 597,000 in 2020 and is expected to increase to 1.7 million by 2050 (Alzheimer Society, 2022). Total health system and out-of-pocket costs for dementia care in Canada are currently estimated to be \$10 billion and the rising number of individuals living with dementia will have a substantial effect on health and social care systems (Alzheimer Society, 2022). Effective treatment options for dementia are limited, placing an emphasis on the importance of prevention (Alzheimer Society, 2022; WHO, 2023). The effectiveness of prevention strategies, however, is dependent on identifying and understanding potentially modifiable risk factors associated with dementia as well as their prevalence and distribution within a population (Alzheimer Society, 2022).

The 2020 report of the Lancet Commission on dementia prevention, intervention, and care identified 12 potentially modifiable risk factors (Livingston et al., 2020): less education, hearing loss, traumatic brain injury (TBI),

hypertension, excessive alcohol consumption, obesity, smoking, depression, social isolation, physical inactivity, air pollution, and diabetes. They estimated that each risk factor accounted individually for between 2.0% and 22.0% of dementia cases and together accounted for 40% of dementia cases worldwide (Livingston et al., 2020). The distribution of modifiable risk factors for dementia has been examined in detail in other countries to inform national and regional public health strategy (Lee et al., 2022; Livingston et al., 2020; Vergara et al., 2022). However, research using national Canadian data has been lacking despite the importance of research on modifiable risk factors being highlighted in Canada's national dementia strategy (Public Health Agency of Canada, 2019). In addition, research linking individual and area-level socio-economic deprivation with increased dementia incidence has suggested dementia prevention efforts would be particularly effective if targeted to households with fewer socio-economic resources (Klee et al., 2023). However, little work has examined how the distribution of modifiable risk factors for dementia varies by income in Canada.

The objective of this study is to examine the prevalence and impact of potentially modifiable risk factors for dementia in Canada based on data from the Canadian Longitudinal Study on Aging (CLSA). Our specific aims are to estimate (1) the prevalence of 12 potentially

modifiable risk factors for all-cause dementia in the Canadian middle-aged and older adult population, and (2) the proportion of dementia cases attributable to each risk factor, i.e. the population attributable fraction (PAF), alone and in combination. Secondly, we will examine how the distribution of these risk factors varies by household income.

Methods

Setting and study type

We conducted a cross-sectional study of middle-aged and older adults in Canada.

Data sources

The Canadian Longitudinal Study on Aging (CLSA) is a national, stratified sample of 51,338 Canadian residents between the ages of 45 and 85 at baseline recruitment (2011–2015). Participants in CLSA were recruited via three sampling frames (the Canadian Community Health Survey, random-digit dialing, and provincial health registries). The CLSA comprised two cohorts, Tracking and Comprehensive. This study uses data only from the Comprehensive cohort as the calculation of all 12 risk factors of interest is not possible on the Tracking cohort. Comprehensive cohort participants ($n = 30,097$ at baseline) were randomly selected within proximity of 11 data collection sites in seven provinces (Newfoundland and Labrador, Nova Scotia, Ontario, Quebec, Manitoba, Alberta, British Columbia) with data collected in-person during home interviews and at the data collection sites. The methodology and sampling strategy of the CLSA has been fully described elsewhere (Raina et al., 2009, 2019). Air quality data were collected and provided by the Canadian Urban Environmental Health Research Consortium and Health Canada and linked to participants by postal code and interview date.

Study participants

We included all Comprehensive cohort participants at baseline (2011–2015) and follow-up 1 (2015–2018). At the time of recruitment, CLSA participants had to be able to complete interviews in English or French and be able to participate in the study on their own. Residents of the three Canadian territories, First Nations reserves, or institutions; full-time members of the Canadian Armed Forces; and individuals unable to consent due to cognitive impairment were excluded from baseline recruitment.

Participant characteristics

We extracted self-reported data on demographics and income. Income measures were based on total household income and categorized as implemented in the CLSA: less than \$20,000, \$20,000–\$49,999, \$50,000–\$99,999, \$100,000–\$149,999 and \$150,000 or higher.

Modifiable risk factors for dementia

The definitions for the 12 modifiable risk factors associated with all-cause dementia were guided where possible by the 2020 Lancet Commission Report, other published work on modifiable risk factors for dementia, and the 2020 Canadian federal air quality guidelines (Environment & Climate Change Canada, 2021; Lee et al., 2022; Livingston et al., 2020). The specific definitions for each risk factor are listed in Table 1. A comparison of these definitions with similar studies can be found in Supplemental Table S1 and justifications for these definitions are provided in Supplemental Table S2. All definitions except for air quality, hearing loss, and obesity were defined using CLSA self-report interview data. Air quality was based on national land-use regression models and satellite-derived measures (Boys et al., 2014; Kharol et al., 2017; McLinden et al., 2014; van Donkelaar et al., 2019). Hearing loss was calculated using pure tone audiometry. Obesity was calculated using in-person BMI measurements. The relative risks associated with each risk factor were taken from the meta-analysis reported by the Lancet Commission (Livingston et al., 2020).

Statistical analysis

We constructed a descriptive profile of CLSA Comprehensive cohort participants at baseline, summarizing demographics and socio-economic factors. The prevalence and 95% confidence interval of each modifiable risk factor was calculated using sampling weights, which were designed to generalize to the target population of Canadian middle-aged and older adults. CLSA sampling weights were calibrated using age, sex, and education (within geographical regions) (CLSA, 2023). Within each calculation, individuals with missing data were removed on a per factor basis. For 10 of the risk factors, the proportion of missing values was minimal, ranging from 0% to 0.63%. Missingness was somewhat higher for physical inactivity (4.4%) and hearing loss (5.0%).

The crude PAF for each risk factor was estimated by Levin's formula (Levin, 1953). Additionally, following the methodology introduced by Norton et al., we produced weighted PAFs to account for overlap between risk factors (Norton et al., 2014). We generated communality weights by conducting principal component analysis on the tetrachoric

Table 1 Definitions of potentially modifiable risk factors for dementia

Risk factor	Definition
Less education	Less than secondary school graduation
Hearing loss	Pure-tone audiometry: Average loss in better ear > 25 dB measured at 0.5, 1, 2, and 4 kHz
Traumatic brain injury	Lifetime history of head injury resulting in loss of consciousness, loss of memory, or dazedness
Hypertension	Self-reported medical diagnosis of high blood pressure or hypertension
Excessive alcohol consumption	Self-reported weekly frequency of drinking more than 15 alcoholic drinks for males and 10 for females in the past 12 months (1 drink = 13.45 g of pure alcohol)
Obesity	Measured body mass index (BMI) ≥ 30 kg/m ²
Smoking	Self-reported current smoking (at least one cigarette every day for the past 30 days)
Depression	CESD-10 score ≥ 10
Physical inactivity	Self-reported physical activity of less than 75 min of vigorous activity or less than 150 min of combined moderate and vigorous activity per week
Diabetes	Self-reported medical diagnosis of diabetes, borderline diabetes, or high blood sugar
Social isolation	Self-reported frequency of participation in family/friend activities outside of household, religious activities, physical activities with other, educational/cultural activities, association activities, volunteer/charity work, and other recreational activities all less than once per week
Air pollution	Nitrogen dioxide (NO ₂): Annual average ≥ 17 ppb <i>or</i> Fine particle matter (PM _{2.5}): 3-year annual average ≥ 8.8 µg/m ³ <i>or</i> Sulfur dioxide (SO ₂): 3-year annual average ≥ 5 ppb

correlation matrix of the 12 risk factors, extracting only components with eigenvalues > 1. The communality for each risk factor, which represents the degree to which it overlaps the other risk factors, was calculated as the sum of the squared factor loadings. The corresponding weight for each risk factor, representing the factor's uniqueness, was the complement of the communality.

We calculated two combined PAFs with different assumptions regarding the form of interactions between risk factors. The first is a widely used method developed by Norton et al. that multiplicatively combines the individual communality-weighted PAFs (Lee et al., 2022; Livingston et al., 2020; Norton et al., 2014). This method was developed to yield more plausible estimates of PAFs involving multiple risk factors than other methods that multiplicatively combined crude PAFs (Norton et al., 2014; Welberry et al., 2023). The second method of calculating a combined PAF assumes additive interactions between risk factors and has been suggested as an alternative to the communality-weighted multiplicative approach (Welberry et al., 2023). A technical appendix with all formulae can be found in Supplemental Table S3. We derived confidence intervals for all PAFs using a two-step bootstrapping approach to account for uncertainty in both the risk factor prevalence and relative risk estimate (Welberry et al., 2023). In this process we generated 10,000 replicate samples with replacement and within each replicate we parametrically sampled relative risks for each risk factor based on their reported confidence intervals.

We repeated all analyses within household income categories. We reported the prevalence and combined PAFs

within each category of income and plotted the prevalence by income to facilitate comparison. Data were analyzed using the R programming language with the packages “survey” and “psych” (Lumley, 2023; Revelle, 2024).

Sensitivity analysis

We conducted several sensitivity analyses to examine the robustness of our results. First, we compared the prevalence of risk factors across the tracking, comprehensive, and combined cohorts of the CLSA. Next, we repeated our primary analysis using the prevalence estimates from the combined cohort when possible and the comprehensive cohort otherwise. Following this, we examined several alternate definitions for risk factors and compared their prevalence to the primary definitions. Additionally, we repeated the primary analysis at the first follow-up data collection point in CLSA, which was approximately 3 years after baseline (2015–2018). We also repeated the primary analysis using the crude (unweighted) prevalence for each risk factor. Finally, we examined the prevalence and PAF of each risk factor at baseline by sex.

Results

The baseline CLSA Comprehensive cohort included 30,097 participants and was 50.9% female (Table 2). The median age was 62 years with first and third quartiles of 54 and 71 years. Less than 10% of participants (8.1%) resided in rural areas.

Prevalence of risk factors

Using sample-weighted proportions, physical inactivity was the most prevalent risk factor (63.8%; 95% CI, 62.8–64.9) followed by hypertension (32.8%; 31.7–33.8), obesity (30.8%; 29.7–31.8), and traumatic brain injury (TBI) (23.9%; 22.9–24.8) (Table 3). Smoking (9.6%; 8.9–10.3), excessive alcohol consumption (11.4%; 10.8–12.0), and hearing loss (15.7%; 14.9–16.5) were the least common risk factors in the Canadian middle-aged and older adult population.

Populational attributable fractions

Physical inactivity (19.9%; 9.2–29.6), hypertension (16.6%, 4.9–28.9), and TBI (16.7%, 11.4–22.3) had the highest crude PAFs (Table 3). Conversely, air pollution (1.7%; 1.3–2.1), excessive alcohol consumption (2.0%; 0.7–3.4), and smoking (5.4%; 1.5–10.3) had the lowest crude PAFs. In the principal component analysis, we extracted five factors with eigenvalues > 1, representing 60% of the total variation. The communality values for each risk factor varied from 42% to 82%. Accounting for overlap between risk factors, the weighted PAFs indicated that physical inactivity (11.6%; 5.4–17.1), depression (7.7%; 5.0–10.7), and hypertension (6.0%; 1.8–10.4) were associated with the most dementia cases while excessive alcohol consumption (0.4%; 0.1–0.6), air pollution (0.8%; 0.6–1.0), and smoking (2.1%; 0.6–4.0) were responsible for the fewest cases. The combined PAFs were 43.4% (37.4–49.0) by the weighted multiplicative method, and 60.9% (55.7–65.5) by the additive method.

Variation in risk by income

The lowest income group exhibited the highest point prevalence for all risk factors except excessive alcohol consumption and hypertension (Fig. 1; Table 4). There was a consistent gradient of decreasing prevalence and both PAFs (both crude and weighted) across increasing income categories for 9 of the 12 risk factors: less education, hearing loss, obesity, smoking, depression, social isolation, physical inactivity, diabetes, and air pollution (Fig. 1). By the weighted multiplicative combined PAF, the 12 risk factors accounted for 58.7% of dementia cases in the lowest income group but only 31.8% in the highest income group. The combined additive PAF was 71.3% in the lowest income group and 51.2% in the highest group (Table 4).

Sensitivity analyses

The prevalence of most risk factors was similar across cohorts. A notable exception was air pollution, the prevalence of which was markedly lower in the more rural tracking cohort (12.6%) compared to the comprehensive cohort

Table 2 Characteristics of CLSA Comprehensive cohort at baseline

Characteristic	CLSA Comprehensive cohort (<i>n</i> = 30,097)
Sociodemographic	
Sex, female	15,320 (50.9%)
Age, yrs, median (Q1, Q3)	62 (54, 71)
Age, yrs	
45–54	7595 (25.2%)
55–64	9856 (32.7%)
65–74	7362 (24.5%)
75 +	5284 (17.6%)
Rural resident	2424 (8.1%)
Cultural/racial background	
White only	28,372 (94.4%)
Black only	221 (0.7%)
Chinese only	212 (0.7%)
South Asian only	271 (0.9%)
Multiple	447 (1.5%)
Other	544 (1.8%)
Total household income (CAD)	
< \$19,999	1566 (5.6%)
\$20,000–\$49,999	6360 (22.6%)
\$50,000–\$99,999	9907 (35.2%)
\$100,000–\$149,999	5524 (19.6%)
≥ \$150,000	4799 (17.0%)

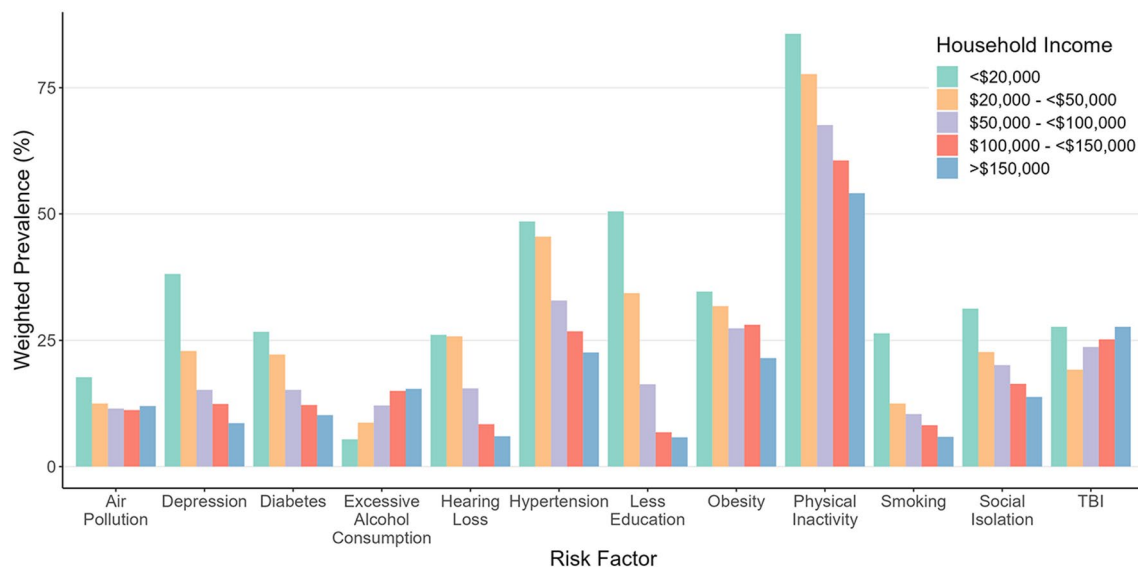
(19.0%) (Supplemental Table S4). Repeating the analysis using prevalence estimates from the combined cohort when possible, and from the comprehensive cohort otherwise, yielded similar results to the main analysis (Supplemental Table S5). The top three risk factors by prevalence, crude PAF, and weighted PAF were all the same as the primary analysis. Overall weighted PAFs were also very similar.

Some of the alternate definitions markedly changed prevalence estimates (Supplemental Tables S6 and S7). For example, considering unilateral rather than bilateral hearing loss doubled the prevalence of the risk factor (15.7% to 32.8%). Applying stricter alcohol guidelines (> 6 drinks a week) nearly tripled excessive alcohol consumption (11.4% to 27.6%). Applying the much stricter WHO 2021 Air Quality Guidelines captured nearly 90% of participants. We believe however that we have chosen the most reasonable definitions for our primary analysis and have provided justification for them in Supplemental Table S2.

The risk factor prevalences and PAFs were broadly similar at follow-up 1 compared to baseline (Table 5). The prevalence of air pollution dropped from 19.0% at baseline to 13.1% at follow-up 1. Physical inactivity was still the most prevalent risk factor with the highest crude and weighted PAFs. As with the baseline analysis, hypertension and TBI had the next highest crude PAFs, followed closely by obesity. The combined PAFs were also similar between baseline and follow-up 1.

Table 3 The weighted prevalence, relative risk, communality, and population attributable fractions (PAF) of dementia risk factors in the CLSA comprehensive cohort at baseline (2011–2015, $n = 30,097$)

Risk factor	Prevalence, % (95% CI)	Relative risk (95% CI)	Communa- lity %	Crude, PAF, %	Weighted, PAF, %
Less education	17.4 (16.1 – 18.6)	1.59 (1.26 – 2.01)	54	9.3 (4.3, 15.0)	4.3 (2.0, 6.9)
Hearing loss	15.7 (14.9 – 16.5)	1.94 (1.38 – 2.73)	74	12.9 (5.7, 21.5)	3.3 (1.5, 5.6)
Traumatic brain injury (TBI)	23.9 (22.9 – 24.8)	1.84 (1.54 – 2.20)	71	16.7 (11.4, 22.3)	4.8 (3.3, 6.5)
Hypertension	32.8 (31.7 – 33.8)	1.61 (1.16 – 2.24)	64	16.6 (4.9, 28.9)	6.0 (1.8, 10.4)
Excessive alcohol consumption	11.4 (10.8 – 12.0)	1.18 (1.06 – 1.31)	82	2.0 (0.7, 3.4)	0.4 (0.1, 0.6)
Obesity	30.8 (29.7 – 31.8)	1.60 (1.34 – 1.92)	64	15.6 (9.4, 22.1)	5.6 (3.4, 8.0)
Smoking	9.6 (8.9 – 10.3)	1.59 (1.15 – 2.20)	61	5.4 (1.5, 10.3)	2.1 (0.6, 4.0)
Depression	18.5 (17.6 – 19.4)	1.90 (1.55 – 2.33)	46	14.3 (9.3, 19.8)	7.7 (5.0, 10.7)
Social isolation	17.5 (16.6 – 18.5)	1.57 (1.32 – 1.85)	48	9.1 (5.4, 13.0)	4.7 (2.8, 6.8)
Physical inactivity	63.8 (62.8 – 64.9)	1.39 (1.16 – 1.67)	42	19.9 (9.2, 29.6)	11.6 (5.4, 17.1)
Diabetes	17.4 (16.5 – 18.2)	1.54 (1.33 – 1.79)	59	8.6 (5.4, 12.1)	3.5 (2.2, 4.9)
Air pollution	19.0 (18.0 – 20.0)	1.09 (1.07 – 1.11)	52	1.7 (1.3, 2.1)	0.8 (0.6, 1.0)
Combined factors	Combined PAF, % (95% CI)				
Weighted multiplicative	43.4 (37.3, 49.0)				
Additive	60.9 (55.7, 65.5)				

**Fig. 1** Weighted prevalence of dementia risk factors by household income in CLSA comprehensive cohort at baseline. (1) Median Canadian household income in 2012 was \$59,700 for all family units,

\$82,100 for economic families of two persons or more, and \$29,800 for unattached individuals

Avoiding the use of sampling weights noticeably lowered the prevalence estimates of low education, smoking, and air pollution (Supplemental Table S8). However, the top three risk factors by prevalence and PAF remained the same regardless of weighting, and the multiplicative weighted combined PAF (40.5%) and additive combined

PAF (59.2%) were also similar. Analysis stratified by sex also yielded largely similar results (Supplemental Table S9). Males had a higher prevalence of TBI (29.4% vs. 18.8%) and social isolation (20.2% vs. 15.1%) while females had a higher prevalence of depression (21.7% vs 15.1%).

Table 4 The weighted prevalence and combined population attributable fractions (PAFs) for dementia risk factors by income group in the CLSA comprehensive cohort at baseline

Risk factor	Household income				
	< \$20,000	\$20,000–< \$50,000	\$50,000–< \$100,000	\$100,000–< \$150,000	≥ \$150,000
	Prevalence, %	Prevalence, %	Prevalence, %	Prevalence, %	Prevalence, %
Less education	47.3	32.6	13.3	7.2	3.4
Hearing loss	26.1	25.8	15.5	8.4	6.0
TBI	27.7	19.2	23.7	25.2	27.7
Hypertension	39.6	41.5	32.7	26.6	23.9
Excessive alcohol consumption	7.0	10.4	11.3	12.2	14.8
Obesity	42.2	33.5	31.8	28.1	23.9
Smoking	23.6	13.5	8.8	7.7	3.5
Depression	48.3	23.5	16.7	11.7	8.7
Social isolation	26.9	21.1	17.4	15.9	12.2
Physical inactivity	79.5	74.2	64.0	57.5	48.1
Diabetes	28.6	21.9	17.3	13.5	10.2
Air pollution	37.8	27.4	17.9	12.6	12.0
Combined factors	Combined PAF, %				
Weighted multiplicative	58.7	50.0	42.2	36.8	31.8
Additive	71.3	65.7	59.9	55.3	51.2

Discussion

We examined the prevalence and population attributable fractions of 12 potentially modifiable risk factors for dementia in the Canadian middle-aged and older adult population using data from the Canadian Longitudinal Study on Aging. We found that the prevalence of risk

factors varied between 9.6% and 63.8%, with physical inactivity the most prevalent risk factor and smoking the least prevalent. Depending on assumptions regarding the interaction of risk factors, we estimated that between 43.4% and 60.9% of dementia cases among Canadian middle-aged and older adults could be attributed to these 12 risk factors. Nine of the 12 risk factors demonstrated

Table 5 The weighted prevalence, relative risk, communality, and population attributable fractions (PAF) of dementia risk factors in the CLSA comprehensive cohort at follow-up 1 (2015–2018, $n = 27,768$)

Risk factor	Prevalence, % (95% CI)	Relative risk, (95% CI)	Communa- lity %	Crude PAF, %	Weighted PAF, %
Less education	16.0 (14.7, 17.3)	1.59 (1.26 – 2.01)	60	8.6 (4.0, 13.9)	3.4 (1.6, 5.6)
Hearing loss	16.2 (15.4, 17.1)	1.94 (1.38 – 2.73)	72	13.2 (5.7, 21.7)	3.7 (1.6, 6.1)
TBI	24.4 (23.4, 25.5)	1.84 (1.54 – 2.20)	80	17.0 (11.7, 22.6)	3.4 (2.3, 4.5)
Hypertension	34.7 (33.7, 35.8)	1.61 (1.16 – 2.24)	66	17.5 (5.0, 30.0)	5.9 (1.7, 10.2)
Excessive alcohol consumption	11.4 (10.8, 12.1)	1.18 (1.06 – 1.31)	75	2.0 (0.7, 3.5)	0.5 (0.2, 0.9)
Obesity	31.3 (30.2, 32.5)	1.60 (1.34 – 1.92)	57	15.8 (9.5, 22.3)	6.8 (4.1, 9.6)
Smoking	7.7 (7.1, 8.5)	1.59 (1.15 – 2.20)	61	4.4 (1.0, 8.5)	1.7 (0.4, 3.3)
Depression	16.0 (15.1, 17.0)	1.90 (1.55 – 2.33)	45	12.6 (8.0, 17.5)	6.9 (4.4, 9.6)
Social isolation	17.7 (16.7, 18.7)	1.57 (1.32 – 1.85)	57	9.2 (5.5, 13.2)	3.9 (2.4, 5.7)
Physical inactivity	63.7 (62.6, 64.8)	1.39 (1.16 – 1.67)	57	19.8 (9.1, 30.0)	8.5 (3.9, 12.9)
Diabetes	19.4 (18.6, 20.4)	1.54 (1.33 – 1.79)	59	9.5 (6.0, 13.3)	3.9 (2.4, 5.5)
Air pollution	13.1 (12.1, 14.1)	1.09 (1.07 – 1.11)	31	1.2 (0.9, 1.4)	0.8 (0.6, 1.0)
Combined factors	Combined PAF, % (95% CI)				
Weighted multiplicative	40.1 (34.5, 45.2)				
Additive	60.8 (55.5, 65.4)				

clear gradients across income categories, with higher prevalences of risk factors at lower incomes.

Our work employs a similar methodology to other global and regional studies (Lee et al., 2022; Livingston et al., 2020; Vergara et al., 2022); however, a novel aspect of our work is that we were able to perform all calculations, except relative risk estimation, within a single national sample. We also used a bootstrapping approach to calculate confidence intervals for PAFs that account for variation in both the prevalence of risk factors and in the relative risks, an improvement over other studies that accounted only for variation within the relative risks (Lee et al., 2022) or did not calculate confidence intervals (Livingston et al., 2020). Despite using a similar methodology, the prevalence and PAFs produced in this study exhibit key differences from those in the 2020 Lancet Commission report. While we found the risk factors responsible for the most dementia cases in Canada to be physical inactivity (wPAF = 11.6%), depression (7.7%), and hypertension (6.0%), the Lancet Commission identified hearing loss (8.2%), less education (7.1%), and smoking (5.2%) as the most impactful risk factors. Differences in prevalence estimates, a result of different target populations and possibly in the definition of risk factors, are largely responsible for the differences in the weighted PAFs. For example, physical inactivity had just an 11.0% prevalence in the Lancet Commission report, compared to 63.8% in our study and 62.8% in a similar study of adults in the United States (Lee et al., 2022). Other Canadian research supports the high prevalence of physical inactivity we measured in this study (Colley et al., 2023). The notable differences in individual PAFs between our study and other international and regional studies are meaningful as they indicate that there are important regional differences in risk factor distribution that can guide the composition and focus of national dementia strategies.

Despite the difference in individual PAFs, the combined PAFs we estimated were similar to those from studies conducted in other jurisdictions. The weighted multiplicative PAF for the 12 risk factors from our study (43%) was highly similar to that of Livingston (Livingston et al., 2017) (40%) and Lee (41%). Additionally, our additive combined PAF (61%) is similar to a global estimate (56%) (Welberry et al., 2023). The differences between the magnitude of the weighted multiplicative and additive combined PAFs are meaningful and ultimately rooted in strong assumptions of the form of interactivity of risk factors within individuals. Interactions between multiple risk factors for dementia are highly complex and the form of the interactions likely varies by each distinct set of risk factors present (Li et al., 2023; O'Donoghue et al., 2016; Welberry et al., 2023). While we believe these combined PAF estimates are useful benchmarks, we have limited evidence on how multiple modifiable

risk factors for dementia interact with each other, and more research on dementia risk factor interaction could improve future estimation of their combined impact.

The results of this study can be used to inform dementia prevention strategies in Canada. Owing to its high prevalence, physical inactivity had the largest PAF, both crude and weighted. Depression had the second highest weighted PAF, a result of its high relative risk and relatively lower correlation with other risk factors. Depression and dementia have a complex and likely bidirectional relationship, but consensus is that depression is an independent risk factor for dementia that could be targeted for intervention (Deckers et al., 2015). Hypertension had the third highest weighted PAF, although this was not meaningfully different from obesity, which had similar prevalence, relative risk, and communality as hypertension. While our study highlights the risk factors with the greatest impact overall, effective prevention requires a combination of individually tailored and population-level interventions. Evidence supports the effectiveness of individualized, multi-domain interventions to reduce risk factors for dementia (Isaacson et al., 2019). Yet many of the most impactful risk factors have structural etiologies which call for a population-level approach to address (Walsh et al., 2023).

Notably, we demonstrated gradients of increasing risk with decreasing income across 9 of the 12 modifiable risk factors. This is concordant with observations of a higher incidence of dementia among lower income households and higher combined PAFs of risk factors in lower income countries (Korhonen et al., 2020; Suemoto et al., 2023). The etiology of the relationship between socio-economic status and dementia risk is likely multifaceted and related to multiple behavioural, social, cultural, occupational, and environmental mechanisms. People more likely to be living below low-income thresholds include recent immigrants, Indigenous people, people living with disabilities, and unattached individuals (Employment & Social Development Canada, 2016). The consistently higher prevalence of dementia risk factors in lower income groups suggests that social-economic disparities in dementia incidence are unlikely to be resolved by a prevention focus on single factors. Policies to address the socio-economic disparities directly may be more effective. Future work should further investigate the relationship between socio-economic status and dementia, particularly the degree to which addressing material deprivation may reduce the prevalence of dementia risk factors and incidence of dementia.

Limitations

Our study is primarily based on self-reported data, which while common in this field of research may introduce recall and social desirability biases. Compared to the target population, the comprehensive CLSA cohort was whiter, healthier, and wealthier (Raina et al., 2019). Our use of sampling weights calibrated by education partially addresses this

limitation. Our sensitivity analyses suggest that integrating the CLSA tracking cohort into our analysis to improve representativeness does not meaningfully alter our results. Nevertheless, the representation gap remains a limitation of our research. Certain groups that are underrepresented, such as racial minorities, or excluded, such as Indigenous people on-reserve, likely have higher prevalences of dementia risk factors (MacDonald et al., 2015). Additionally, the underrepresentation of lower income groups suggests some caution is warranted in interpreting the prevalences of risk factors by income category.

Individuals with cognitive impairment significant enough to prevent them from giving informed consent were excluded from the CLSA at baseline, and this may have resulted in underestimation of the prevalence of some risk factors. With respect to the PAF calculations, the relative risks used were calculated from an international systematic review and are not specific to Canada. While we strove to align our risk factor definitions with previous work, the Lancet Commission report from which the relative risks were taken used data from multiple sources and did not always have a single, specific definition. Differences in risk factor definitions would lead to meaningful differences in findings for some risk factors. The greatest uncertainty is with air pollution, which involves multiple, independent exposures that are inherently continuous rather than dichotomous. Future research could focus on more precisely estimating the contribution of air pollutants to the incidence of dementia in Canada.

Conclusion

We examined the prevalence and population attributable fractions of 12 potentially modifiable risk factors for dementia and estimated that they accounted for between 43.3% and 60.9% of dementia cases among middle-aged and older adults in Canada. Accounting for overlap between risk factors, the factors with the highest overall impact on dementia cases were physical inactivity, depression, and hypertension. The differences in the impact of individual risk factors observed between this study and other international and regional studies highlight the importance of tailoring national and regional dementia strategies to the local distribution of risk factors.

Contributions to knowledge

What does this study add to existing knowledge?

- We have estimated the individual and combined contributions of 12 modifiable risk factors to the incidence of dementia, using national Canadian data.

- We have identified the risk factors with the greatest impact on the number of dementia cases in Canada.
- We have identified a clear association between increasing burden of risk factors for dementia and decreasing income.

What are the key implications for public health interventions, practice, or policy?

- A large proportion of dementia cases in Canada are due to modifiable risk factors that could be targets of population-level interventions.
- The most impactful risk factors are not necessarily the same as those identified in other international and regional studies, underscoring the importance of tailoring a Canadian dementia strategy to Canadian data.
- There is a clear need to address socio-economic disparities in accumulation of dementia risk factors.

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Author contributions AJ, AC, LG, and PR conceived the work and developed the design in consultation with the other authors. YD and AJ performed the data analysis. YD wrote the initial draft of the paper. All authors contributed to the interpretation of the data and revised the draft for important intellectual content. All authors approve the final version to be published and agree to be accountable for all aspects of the work.

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Availability of data and material Data are available from the Canadian Longitudinal Study on Aging (www.clsa-elcv.ca) for researchers who meet the criteria for access to de-identified CLSA data.

Code availability Code can be made available upon request from the corresponding author.

Declarations

Conflict of interest The authors declare they have no competing interests.

Ethics This study received ethics approval from the Hamilton Integrated Research Ethics Board (15408).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication NA.

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